



JNTUA College of Engineering (Autonomous) Ananthapuramu
Department of Computer Science and Engineering
MCA (R20)

Program Elective:3

Design Patterns

SEMESTER - IV

L-T-P-C: 3-0-0-3

Course Objectives:

- Understand the concept of Design patterns and its importance.
- Understand the behavioral knowledge of the problem and solutions.
- Relate the Creational, Structural, behavioral Design patterns.
- Apply the suitable design patterns to refine the basic design for given context.

Course Outcomes (COs): At the end of the course, student will be able to

Course Outcomes		Knowledge Level (K)#
CO1	Illustrate the appropriate design patterns to solve object-oriented design problems.	K2
CO2	Apply structural patterns to solve design problems.	K3
CO3	Evaluate the design solutions by using behavioral patterns.	K5
CO4	Develop design solutions using creational patterns	K6
CO5	Demonstrate about Advanced Patterns like Pattern Catalogs	K3

#Based on suggested Revised BTL

Mapping of course outcomes with program outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1		3					2			
CO2		3					1			
CO3					2		1			
CO4								2		
CO5			1					2		

(Levels of Correlation: 1-low, 2-medium 3-high)

Unit-I:

Introduction: History and Origin of Patterns, Design Patterns in MVC, Describing Design Patterns, How Design Patterns Solve Design Problems, selecting a Design Pattern, Using a Design Pattern

Unit-II:

Design Patterns-1: Creational, Abstract Factory-Builder, Factory Method, Prototype-Singleton

Unit- III:

Design Patterns-2: Structural Patterns: Adapter, Bridge, Composite, Decorator, Façade, Flyweight, Proxy

Unit-IV:

Design Patterns-3: Behavioural Patterns, Chain of Responsibility, Command-Interpreter, Iterator- Mediator, Memento, Observer, State, Strategy, Template Method, Visitor

Unit-V:

Advanced Patterns: Pattern Catalogs and Writing Patterns, Patterns and Case Study: Designing a Document Editor Anti-Patterns - Case Studies in UML and CORBA, Pattern Community.

Text Books:

1. Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides, Design patterns: Elements of Reusable object-oriented software, Addison-Wesley, 1995.
2. James W Cooper, Java Design Patterns - A Tutorial, Addison-Wesley, 52000.

Reference Books:

1. Craig Larman, Applying UML and Patterns: An Introduction to object-Oriented Analysis and Design and iterative development, 3rd Edition, Pearson, 2005.
2. Thomas J Mowbray and Raphael Malveau, CORBA and Design Patterns, John Wiley, 1997.
3. William J Brown, Anti-Patterns: Refactoring Software, Architectures and Projects in Crisis, John Wiley, 1998.



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Program Elective-4

Big Data Analytics

SEMESTER - IV

L-T-P-C: 3-0-0-3

Course Objectives:

- To know the fundamental concepts of big data and analytics.
- To explore tools and practices for working with bigdata
- To learn about stream computing.
- To know about the research that requires the integration of large amounts of data.

Course Outcomes (COs): At the end of the course, student will be able to

Course Outcomes		Knowledge Level (K)#
CO1	Identify the need-based tools, viz., Pig and Hive and to handle and formulate an effective strategy to implement a successful Data analytics project	K3
CO2	Organize the existing technologies and the need of distributed files systems to analyze the big data	K3
CO3	To Discuss the cluster and classification techniques	K5
CO4	Analyze the concepts of stream memory and spark models.	K4
CO5	Explain the use of NoSQL database in data analytics.	K5

#Based on suggested Revised BTL

Mapping of course outcomes with program outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1			3					1		
CO2	1				2					
CO3					2					
CO4								3		
CO5			3							

(Levels of Correlation: 1-low, 2-medium 3-high)

UNIT-I:

Introduction to Big Data- Evolution of Big data, Best Practices for Big data Analytics, Big data characteristics, Validating, The Promotion of the Value of Big Data, Big Data Use Cases, Characteristics of Big Data Applications, Perception and Quantification of Value, Understanding Big Data Storage, A General Overview of Hadoop Architecture, Performance Architecture, HDFS, MapReduce and YARN, Map Reduce Programming Model

UNIT-II:

Frameworks-Applications on Big Data Using Pig and Hive, Data

processing operators in Pig, Hive services, HiveQL, Querying Data in Hive, fundamentals of HBase and Zoo Keeper, IBM InfoSphere Big Insights and Streams

UNIT-III:

Clustering and Classification-Advanced Analytical Theory and Methods: Overview of Clustering, K-means, Use Cases - Overview of the Method, Determining the Number of Clusters, Diagnostics, Reasons to Choose and Cautions. Classification: Decision Trees, Overview of a Decision Tree, The General Algorithm, Decision Tree Algorithms, Evaluating a Decision Tree, Decision Trees in R, Naïve Bayes, Baye's Theorem, Naïve Bayes Classifier.

UNIT- IV:

Stream Memory and Spark- Introduction to Streams Concepts, Stream Data Model and Architecture, Stream Computing, Sampling Data in a Stream, Filtering Streams, Counting Distinct Elements in a Stream, Introduction to Spark Concept, Spark Architecture and components, spark installation, spark RDD(Resilient Distributed Dataset), spark RDD operations.

UNIT-V:

NOSQL Data Management for Big Data and Visualization- NoSQL Databases: Schema-less Models: Increasing Flexibility for Data Manipulation, Key Value Stores, Document Store, Tabular Stores, Object Data Stores, Graph Databases Hive, Sharding, Hbase, Analyzing big data with twitter, Big data for E-Commerce Big data for blogs, Review of Basic Data Analytic Methods using R.

Text Books:

1. Anand Rajaraman and Jeffrey David Ullman, "Mining of Massive Datasets", Cambridge University Press, 2012.
2. David Loshin, "Big Data Analytics: From Strategic Planning to Enterprise Integration with Tools, Techniques, NoSQL, and Graph", Morgan Kaufmann/El sevier Publishers, 2013.

Reference Books:

1. Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer, 2007.
2. Tom White "Hadoop: The Definitive Guide" Third Edition, O'Reilly Media, 2012.
3. Chris Eaton, Dirk DeRoos, Tom Deutsch, George Lapis, Paul Zikopoulos, "Understanding Big Data: Analytics for Enterprise Class Hadoop and Streaming Data", McGrawHill Publishing, 2012.
4. Bill Franks, "Taming the Big Data Tidal Wave: Finding Opportunities in Huge Data Streams with Advanced Analytics", John Wiley & sons, 2012.
5. Glenn J. Myatt, "Making Sense of Data", John Wiley & Sons, 2007.
6. Pete Warden, "Big Data Glossary", O'Reilly, 2011.
7. Jiawei Han, Micheline Kamber "Data Mining Concepts and Techniques", 2nd Edition, Elsevier, Reprinted 2008.